

# Anas Barakat

Last update: October 7th 2026

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## RESEARCH OVERVIEW

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Optimization and learning for strategic and sequential decision making beyond expected rewards, under limited feedback:

- **Policy optimization beyond expected additive rewards:** general-utility RL and LLM post-training.
- **Learning and optimization in structured games:** multi-agent RL, Markov games with general utilities, learning dynamics, and equilibrium learning.
- **Online learning beyond convexity under limited feedback:** online learning, bandit and preference feedback, nonconvex optimization, and hidden convexity.

## ACADEMIC APPOINTMENTS

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### Singapore University of Technology and Design

Postdoctoral Research Fellow

Oct. 2024 -

Hosts: Prof. Georgios Piliouras and Prof. Antonios Varvitsiotis

### ETH Zurich, Department of Computer Science

Foundations of Data Science Postdoctoral Fellow

Feb. 2022 - Aug. 2024

Host: Prof. Niao He

## EDUCATION

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### Institut Polytechnique de Paris, Télécom Paris, Paris, France

Ph.D. in Applied Mathematics and Computer Science

Fall 2018 - Fall 2021

*Advisors:* Prof. Pascal Bianchi and Prof. Walid Hachem.

*Thesis:* Contributions to non-convex stochastic optimization and reinforcement learning

*Committee:* Profs. Vivek S. Borkar, Sébastien Gadat, Robert M. Gower, Niao He, and Edouard Pauwels

### Université Paris Saclay, Paris, France

M.Sc. in Data Science (with highest honors)

Fall 2017 - Summer 2018

### Télécom Paris, Paris, France

M.Sc. in Applied Mathematics and Computer Science

Fall 2015 - Summer 2018

## SELECTED PUBLICATIONS

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1. **Anas Barakat**. *Learning in Continuous Games from Pairwise Preference Feedback*. Preprint, 2026.
2. **Anas Barakat**, Souradip Chakraborty, Khushbu Pahwa, Amrit Singh Bedi. *Why Pass@k Optimization Can Degrade Pass@1: Prompt Interference in LLM Post-training*. NeurIPS 2026.
3. **Anas Barakat**, Andreas Kontogiannis, Vasilis Pollatos, Ioannis Panageas, Antonios Varvitsiotis. *Online Learning on Hidden-Convex Losses via Algorithmic Equivalence: Optimal Regret, Geometric Barrier, and Bandit Feedback*. NeurIPS 2026.
4. **Anas Barakat**, John Lazarsfeld, Georgios Piliouras, Antonios Varvitsiotis. *Online Learning in Stabilized Linear Dynamical Games with Adversarial Disturbances*. NeurIPS 2026.
5. **Anas Barakat**, Ioannis Panageas, Antonios Varvitsiotis. *Convex Markov Games and Beyond: New Proof of Existence, Characterization and Learning Algorithms for Nash Equilibria*. AISTATS 2026.

6. **Anas Barakat**, Souradip Chakraborty, Peihong Yu, Pratap Tokekar, Amrit Singh Bedi. *On the Global Optimality of Policy Gradient Methods in General Utility Reinforcement Learning*. NeurIPS 2025.
7. **Anas Barakat**, Ilyas Fatkhullin, Niao He. *Reinforcement Learning with General Utilities: Simpler Variance Reduction and Large State-Action Space*. ICML 2023.

## PUBLICATIONS & PREPRINTS

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Names of student co-authors I have previously supervised and mentored are underlined.

### Preprints

- (1) **Anas Barakat**. *Learning in Continuous Games from Pairwise Preference Feedback*. Preprint, 2026.
- (2) John Lazarsfeld, **Anas Barakat**, Jacob Abernethy, Vidya Muthukumar, Georgios Piliouras, Andre Wibisono, Juba Ziani. *Linear Last-Iterate Convergence Rates for Optimistic Learning In Harmonic Games*. Preprint, 2026.

### Refereed Journal Publications

- (J1) Olivier Lepel, **Anas Barakat**. *Policy Gradients for Cumulative Prospect Theory in Reinforcement Learning*. Transactions on Machine Learning Research, 2026.
- (J2) **Anas Barakat**, Wayne Lin, John Lazarsfeld, Antonios Varvitsiotis. *Optimistic Online Learning in Symmetric Cone Games*. Transactions on Machine Learning Research, 2026.
- (J3) Jiduan Wu, **Anas Barakat**, Ilyas Fatkhullin, Niao He. *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last Iterate Convergence*. SIAM Journal on Control and Optimization, 2025.
- (J4) **Anas Barakat**, Pascal Bianchi, Walid Hachem, Sholom Schechtman. *Stochastic optimization with momentum: convergence, fluctuations, and traps avoidance*. Electronic Journal of Statistics 15 (2), 3892-3947, 2021. <https://doi.org/10.1214/21-EJS1880>
- (J5) **Anas Barakat**, Pascal Bianchi. *Convergence and Dynamical Behavior of the Adam Algorithm for Non-Convex Stochastic Optimization*. SIAM Journal on Optimization 31 (1), 244-274, 2021.

### Refereed Conference Publications

- (C1) **Anas Barakat**, Souradip Chakraborty, Khushbu Pahwa, Amrit Singh Bedi. *Why Pass@k Optimization Can Degrade Pass@1: Prompt Interference in LLM Post-training*. Advances in Neural Information Processing Systems (NeurIPS 2026).
- (C2) **Anas Barakat**, Andreas Kontogiannis, Vasilis Pollatos, Ioannis Panageas, Antonios Varvitsiotis. *Online Learning on Hidden-Convex Losses via Algorithmic Equivalence: Optimal Regret, Geometric Barrier, and Bandit Feedback*. Advances in Neural Information Processing Systems (NeurIPS 2026).
- (C3) John Lazarsfeld, **Anas Barakat**, Georgios Piliouras, Antonios Varvitsiotis, Andre Wibisono. *When and Why is Optimistic Multiplicative Weights Slow? The Geometry of Energy Dissipation*. Advances in Neural Information Processing Systems (NeurIPS 2026).
- (C4) **Anas Barakat**, John Lazarsfeld, Georgios Piliouras, Antonios Varvitsiotis. *Online Learning in Stabilized Linear Dynamical Games with Adversarial Disturbances*. Advances in Neural Information Processing Systems (NeurIPS 2026).
- (C5) **Anas Barakat**, Ioannis Panageas, Antonios Varvitsiotis. *Convex Markov Games and Beyond: New Proof of Existence, Characterization and Learning Algorithms for Nash Equilibria*. International Conference on Artificial Intelligence and Statistics (AISTATS 2026).

- (C6) **Anas Barakat**, Souradip Chakraborty, Peihong Yu, Pratap Tokekar, Amrit Singh Bedi. *On the Global Optimality of Policy Gradient Methods in General Utility Reinforcement Learning*. Advances in Neural Information Processing Systems (NeurIPS 2025).
- (C7) Kimion Protopapas, **Anas Barakat**. *Policy Mirror Descent with Lookahead*. Advances in Neural Information Processing Systems (NeurIPS 2024).
- (C8) Pragnya Alatur, **Anas Barakat\***, Niao He. *Independent Policy Mirror Descent for Markov Potential Games: Scaling to Large Number of Players*. IEEE Conference on Decision and Control (CDC 2024). \*Corresponding author.
- (C9) Philip Jordan, **Anas Barakat**, Niao He. *Independent Learning in Constrained Markov Potential Games*. International Conference on Artificial Intelligence and Statistics (AISTATS 2024).
- (C10) Jiduan Wu, **Anas Barakat**, Ilyas Fatkhullin, Niao He. *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity*. IEEE Conference on Decision and Control (CDC 2023).
- (C11) **Anas Barakat**, Ilyas Fatkhullin, Niao He. *Reinforcement Learning with General Utilities: Simpler Variance Reduction and Large State-Action Space*. International Conference on Machine Learning (ICML 2023).
- (C12) Ilyas Fatkhullin, **Anas Barakat**, Anastasia Kireeva, Niao He. *Stochastic Policy Gradient Methods: Improved Sample Complexity for Fisher-non-degenerate Policies*. International Conference on Machine Learning (ICML 2023).
- (C13) **Anas Barakat**, Pascal Bianchi, Julien Lehmann. *Analysis of a Target-Based Actor-Critic Algorithm with Linear Function Approximation*. International Conference on Artificial Intelligence and Statistics (AISTATS 2022).
- (C14) **Anas Barakat**, Pascal Bianchi. *Convergence Rates of a Momentum Algorithm with Bounded Adaptive Step Size for Nonconvex Optimization*. Asian Conference on Machine Learning (ACML 2020).

## AWARDS AND SCHOLARSHIPS

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| • ETH Zurich Foundations of Data Science Postdoctoral Fellowship                                                                                                                                                                                                                                                        | 2022-2024     |
| • DAAD Postdoctoral Networking Tour in AI (Postdoc-NeT-AI) fellow                                                                                                                                                                                                                                                       | Nov. 2023     |
| • Dodu Prize, French Society of Applied and Industrial Mathematics (SMAI)<br>Best communication of a young researcher at the Optimization and Decision annual days<br>Jury: M. Akian, P. Bich, J. Bolte, J-B. Caillaud, S. Gaubert, V. Leclerc, P. Mertikopoulos,<br>F. Santambrogio (president of the jury), H. Zidani | Sep. 9th 2021 |
| • Mines-Télécom Institute (IMT) Ph.D scholarship                                                                                                                                                                                                                                                                        | 2018-2021     |
| • Moroccan Government Merit Scholarship                                                                                                                                                                                                                                                                                 | 2015-2018     |
| • Agency for French Education Abroad (AEFE) “Excellence-Major” scholarship                                                                                                                                                                                                                                              | 2013-2015     |

## RESEARCH TALKS

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| • Invited talk, 5th Symposium on Machine Learning and Dynamical Systems<br>Kyoto University, Kyoto, Japan                                                                  | 10 Feb 2026   |
| • Invited talk, ICCOPT 2025, University of Southern California, Los Angeles, USA                                                                                           | 24 Jul 2025   |
| • Invited talk, Learning Theory and Applications Workshop<br>College of Computing and Data Science, Nanyang Technological University, Singapore<br>Host: Themis Gouleakis. | 29 April 2025 |

- Online invited talk, Finance and RL Talks, FinRL core team & Melwy open AI lab 21 April 2025  
Host: Mostapha Benhenda.
- Online talk, Singapore University of Technology and Design research group 3 May 2024  
Host: Georgios Piliouras.
- Invited talk, Fourth Symposium on Machine Learning and Dynamical Systems 11 July 2024  
Fields Institute, Toronto, Canada.
- Invited talk, ICCOPT 2022, Lehigh University, Bethlehem, USA 28 Jul 2022
- Invited talk (online), 14th CMStatistics International Conference 20 Dec 2021  
Session: “Dynamical systems in machine learning” organized by Anna Korba  
King’s College, London, United Kingdom
- Invited Seminar, *Image, Optimization and Probabilities* research group 15 Oct 2020  
Bordeaux Institute of Mathematics (IMB), Bordeaux, France
- 2nd Symposium on Machine Learning and Dynamical Systems 21 Sep 2020  
Fields Institute for Research in Mathematical Sciences, online
- Mathematical Optimization and Decision (MODE) group days 7 Sep 2020  
French Society of Applied and Industrial Mathematics (SMAI), online
- Mathematics of Optimization and Applications (MOA) annual days 17 Oct 2019  
National Institute of Applied Sciences (INSA), Rennes, France
- 11th OPT Workshop on Optimization for Machine Learning 14 Dec 2019  
Exchange Hotel Vancouver, Vancouver, Canada
- Machine Learning in the Real World workshop 2 Oct 2019  
Criteo, Paris, France
- Junior Conference on Data Science and Engineering 12 Sep 2019  
Centrale Supélec, Gif-sur-Yvette, France
- Francophone colloquium of Signal and Image Processing (GRETSI) 29 Aug 2019  
Lille University, Lille, France

## TEACHING

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- **Singapore University of Technology and Design**, Singapore
  - Co-designer and co-instructor: Online Learning and Learning in Games Fall 2025  
Course website and notes: <https://learning-in-games.github.io>.
- **ETH Zurich**, Zurich, Switzerland
  - Teaching Assistant for ‘Optimization for Data Science’ and ‘Foundations of RL’ Spring 2024
  - Instructor for a lecture about ‘Value-based methods’ Spring 2024
  - Head Teaching Assistant for ‘Foundations of Reinforcement Learning’ (Prof. He) Spring 2023
  - Instructor for 2 lectures for ‘Optimization for Data Science’ (Prof. He) Spring 2023
  - Coordinator for the seminar course ‘Advanced Topics in Machine Learning’ Fall 2022, 2023
- **Télécom Paris**, Paris, France
  - Teaching Assistant for ‘Optimization for Machine Learning’ Fall 2018, 2019, 2020
  - Instructor for discrete Markov chains in ‘Probabilities’ Fall 2018, 2019

– Teaching Assistant for ‘Probabilities’, ‘Statistics’, ‘Machine Learning’

Fall 2018, 2019

- **Ecole Polytechnique**, Paris, France

- Tutor for undergraduate students in Computer Science (about 20 hours) Spring 2017 Design and Analysis of Algorithms; Advanced Programming

## STUDENT MENTORING

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- Olivier Lepel, Master thesis (03/2024-09/2024): *Beyond Expected Returns: A Policy Gradient Algorithm for Cumulative Prospect Theoretic Reinforcement Learning*. TMLR 2026.
- Kimon Protopapas, Master student semester project (09/2023-01/2024): *Policy Mirror Descent with Lookahead*. NeurIPS 2024.
- Philip Jordan, Master thesis (05/2023 - 12/2023): *Independent Learning in Constrained Markov Potential Games*. AISTATS 2024.
- Jiduan Wu, Master thesis (10/2022 - 03/2023): *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last-Iterate Convergence*. IEEE Conference on Decision and Control 2023.
- Harish Rajagopal, Master thesis (03/2022 - 09/2022): *Multistage Step Size Scheduling for Minimax Problems*.
- Julien Lehmann, Master thesis (05/2021 - 10/2021): *Analysis of a Target-Based Actor-Critic Algorithm with Linear Function Approximation*. AISTATS 2022.

## ACADEMIC SERVICE

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- **Area Chair:** ICLR 2027.
- **Conference reviewing:** NeurIPS (2022, 2023, 2025, 2026), ICML (2024, 2025, 2026 Top 25 % - Gold Reviewer), ICLR (2025-2026), AISTATS (2022 Top 10 % reviewer, 2024, 2026), EC 2026.
- **Journal reviewing:** Journal of Machine Learning Research (JMLR), Transactions on Machine Learning Research (TMLR), Mathematical Programming, SIAM Journal on Optimization (SIOPT), Journal of Optimization Theory and Applications (JOTA), IEEE Transactions on Automatic Control (TACON), Stochastic Systems, Systems & Control Letters, Mathematics of Control, Signals, and Systems (MCSS), IEEE Control Systems Letters (IEEE L-CSS).
- **Organization of workshops, summer schools and conferences:**
  - Multi-Agent RL EPFL-ETHZ Summer School 2024 co-organizer (20 000 CHF funding, accepted proposal), supported by Prof. Niao He and Prof. Volkan Cevher.
  - ICCOPT 2022, session organizer and chair: *Policy Gradient and Actor-Critic Methods: Theoretical Analysis and New Opportunities*, in the Optimization for Data Science and ML cluster.
  - Workshop organizer: RL workshop for industrial partners (Airbus Defence & Space, Engie, Idemia, Safran and Valeo) of the Data Science & AI for Digitalized Industry & Services research and teaching chair.